Inventory-Conscious Reinforcement Learning System for Customized E-Commerce Pricing

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Abstract
DRL-PricePro is an advanced deep reinforcement learning platform for personalized dynamic pricing in e-commerce. It incorporates supply constraints by modeling pricing decisions as a constrained Markov Decision Process and utilizes an enhanced Soft Actor-Critic algorithm specifically tailored for e-commerce environments. The framework manages inventory through multi-level constraint handling, blending strict limits with adaptive soft constraints, and personalizes pricing through behavioral customer segmentation. Testing on 2.7 million e-commerce transactions, DRL-PricePro achieved a 12.3% revenue increase over static pricing and a 7.6% improvement over rule-based strategies, with a 98.7% supply constraint satisfaction rate. Case studies confirm robust performance in real applications

Keywords: Deep Reinforcement Learning, Dynamic Pricing, Supply Constraints, Personalization

1. Introduction
1.1. Research Background and Motivation

The e-commerce industry has experienced unprecedented growth in recent years, creating increasingly competitive market environments where pricing strategies play a crucial role in determining business success. Dynamic pricing, as opposed to static pricing mechanisms, allows businesses to adjust prices in real-time based on various factors such as demand fluctuations, competitor pricing, and inventory levels. The financial implications of effective pricing strategies extend beyond immediate revenue generation to broader economic security considerations, as highlighted by Kang et al. in their analysis of financial flow patterns and economic implications [1]. In parallel with the evolution of pricing models, advancements in artificial intelligence and machine learning have created new opportunities for sophisticated price optimization approaches that can process vast amounts of data and extract actionable insights. These technological capabilities are particularly relevant when analyzing nuanced market signals, similar to how Liang et al. developed metrics for detecting subtle sentiment manipulation in financial content [2]. The integration of deep reinforcement learning (DRL) into pricing frameworks represents a significant advancement, enabling systems to learn optimal pricing policies through continuous interaction

with dynamic market environments while maximizing long-term objectives rather than short-term gains.

1.2. Challenges in Dynamic Pricing with Supply Constraints

Despite significant progress in dynamic pricing research, several challenges remain unresolved, particularly when supply constraints are introduced into the equation. Traditional dynamic pricing models often assume unlimited inventory availability, which creates a disconnection from realworld scenarios where stock limitations impact optimal pricing decisions. Wang and Liang highlighted the importance of model interpretability in financial applications [3], which presents a parallel challenge in dynamic pricing systems where transparent decision mechanisms are essential for business stakeholders. Supply constraints add multiple layers of complexity, requiring pricing systems to balance immediate revenue objectives with inventory management goals across different time horizons. Additionally, compliance considerations within pricing systems become increasingly complex in multi-channel environments, reflecting challenges similar to those described by Dong and Zhang in their framework for compliance risk assessment [4]. The personalization aspect of pricing further complicates the landscape, as customer segments exhibit varying price sensitivities and behavioral patterns that must be incorporated into the pricing model while respecting inventory limitations across product categories.

1.3 Research Objectives and Contributions

This paper introduces DRL-PricePro, a novel deep reinforcement learning framework designed specifically for dynamic pricing in e-commerce environments with explicit consideration of supply constraints. The primary objective is to develop a robust, adaptive pricing system that optimizes revenue while efficiently managing inventory limitations. The research leverages recent advances in LSTM-based prediction models, drawing inspiration from the predictive modeling approach demonstrated by Wang et al [5], but adapting these techniques specifically for the e-commerce pricing domain. Our framework incorporates personalization mechanisms that segment customers based on their historical behaviors and preferences, allowing for differentiated pricing strategies across user groups. The feature selection and optimization components of our system build upon machine learning approaches similar to those proposed by Ma et al. [6], but are tailored specifically to identify the most relevant factors influencing pricing decisions under varying supply conditions. Through extensive empirical evaluation, this research demonstrates how deep reinforcement learning can effectively navigate the complex interplay between dynamic pricing and supply constraints while delivering measurable improvements in revenue optimization and inventory efficiency compared to traditional pricing methods.

2. Literature Review

2.1. Dynamic Pricing Strategies in E-commerce

Dynamic pricing has evolved significantly in e-commerce environments, progressing from basic time-based discounting to sophisticated algorithmic approaches that respond to market conditions in real-time. Traditional dynamic pricing methods relied predominantly on historical sales data and rudimentary demand forecasting, but contemporary approaches incorporate multidimensional factors including competitor pricing, customer behavior patterns, and inventory positions. The underlying mechanisms for dynamic pricing vary across implementation contexts, ranging from markdown optimization for seasonal products to auction-based models for limited-availability items. Recent research has focused on enhancing the efficiency and accuracy of dynamic pricing systems through improved anomaly detection capabilities. Li et al. introduced a method for improving database anomaly detection efficiency through sample difficulty estimation, which provides valuable insights for detecting irregular patterns in pricing data streams [7]. These detection capabilities are particularly relevant in dynamic pricing contexts where identifying and responding to market anomalies can create significant competitive advantages. While rule-based dynamic pricing systems continue to see widespread implementation due to their interpretability and straightforward operation, the limitations of these approaches become increasingly apparent in complex market environments where numerous interrelated factors influence optimal price points.

2.2. Machine Learning Approaches for Price Optimization

Machine learning techniques have revolutionized price optimization by enabling businesses to process vast amounts of heterogeneous data and extract actionable pricing insights. Supervised learning approaches utilize historical transaction data to predict customer responses to different price points, while unsupervised methods identify natural customer segments with distinct price sensitivities. The application of deep learning to pricing optimization has gained significant traction, with neural network architectures demonstrating superior performance in capturing complex, non-linear relationships between pricing factors and purchase behaviors. Yu et al. developed a real-time detection system for anomalous trading patterns using generative adversarial networks, which presents methodological parallels to identifying pricing opportunities in volatile markets [8]. Time-series analysis techniques have proven particularly valuable for modeling temporal dependencies in pricing data. Xiao et al. implemented an LSTM-Attention mechanism for detecting anomalous payment behaviors, an approach that can be adapted to identify optimal timing for price adjustments based on historical purchase patterns [9]. Reinforcement learning represents a paradigm shift in pricing optimization, transitioning from reactive to proactive approaches by learning optimal pricing policies through continuous interaction with the market environment without requiring explicit programming of price-response functions.

2.3. Supply Chain Constraints in Pricing Models

The integration of supply chain constraints into pricing models addresses a critical gap in traditional approaches that often treat pricing decisions in isolation from inventory considerations. Limited product availability fundamentally alters the pricing optimization landscape, introducing temporal dependencies where current pricing decisions impact future inventory positions and, consequently, future pricing flexibility. Multi-echelon supply chains present additional complexity, requiring coordinated pricing strategies across interconnected inventory positions. Xiao et al. developed differential privacy mechanisms for data protection [10], which presents relevant methodological considerations for securing sensitive supply chain data used in pricing decisions. The protection of proprietary supply chain information within pricing systems represents an important operational consideration, particularly as pricing algorithms increasingly operate across organizational boundaries. Zhang et al. introduced privacy-preserving feature extraction techniques based on encryption [11], which can be adapted to maintain confidentiality of sensitive supply chain parameters while enabling cross-organizational pricing optimization. Advanced detection algorithms similar to those proposed by Ren et al. for Trojan virus classification using graph convolutional neural networks [12] can be repurposed to identify potential supply chain disruptions that necessitate pricing adjustments. The dynamic interplay between pricing decisions and inventory positions creates a bidirectional relationship where pricing influences demand patterns while inventory constraints simultaneously restrict pricing flexibility.

3. Proposed DRL-PricePro Framework

3.1. System Architecture and Components

The DRL-PricePro framework consists of interconnected components that collectively enable personalized dynamic pricing while respecting supply constraints in e-commerce platforms. The system architecture integrates data processing pipelines, reinforcement learning modules, and constraint management mechanisms within a unified framework. Figure 1 presents the overall architecture of the proposed system, highlighting the information flow between components and the primary decision points in the pricing process.

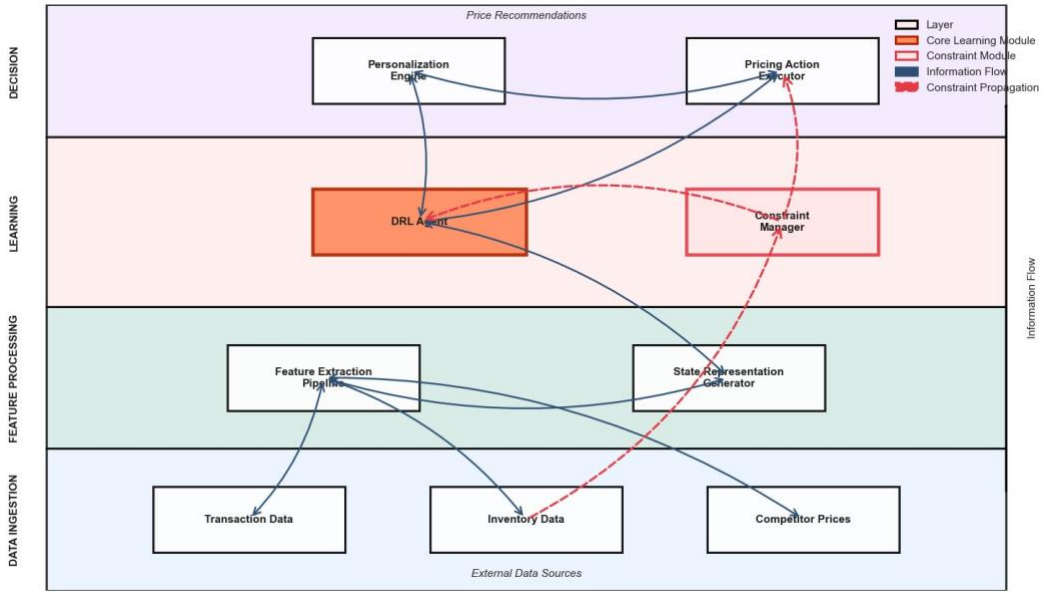

Figure 1: DRL-PricePro System Architecture.

The figure illustrates the multi-layered architecture of the DRL-PricePro framework, consisting of four main layers: data ingestion layer (bottom), feature processing layer, learning layer, and decision layer (top). Bidirectional arrows indicate information flows between components, with special emphasis on the constraint propagation pathways that ensure supply limitations are respected throughout the pricing process. The central learning module is highlighted with a distinctive color to emphasize its role as the core computational engine of the system.

The architecture implements a modular design that facilitates component isolation and system maintainability. Each component addresses specific challenges in dynamic pricing, with particular attention to the technical requirements imposed by supply constraints. The detection methodology employed in the constraint monitoring module bears conceptual similarities to the graph convolutional neural network approach developed by Ren et al. for classification tasks [12], but has been adapted specifically for identifying inventory bottlenecks that may impact pricing decisions. The system components and their respective functions are detailed in Table 1.

Table 1: DRL-PricePro System Components and Functions

Component	**Function**	**Input**	**Output**
Data Ingestion Module	Collects real-time data from multiple sources	Raw transaction data, inventory levels, competitor prices	Structured data streams
Feature Extraction Pipeline	Processes raw data into feature vectors	Structured data streams	Feature vectors
State Representation Generator	Creates state representations for RL agent	Feature vectors, historical states	State tensors

DRL Agent	Learns optimal pricing policies	State tensors, rewards	Pricing actions
Constraint Manager	Enforces supply limitations on pricing decisions	Inventory data, pricing actions	Constrained pricing actions
Personalization Engine	Adapts pricing strategies to customer segments	Customer profiles, historical interactions	Segment-specific adjustments

The data sources leveraged by the system encompass multiple dimensions of the e-commerce environment, providing the DRL agent with comprehensive market visibility. Table 2 outlines the primary data types incorporated into the framework along with their update frequencies and relevance to pricing decisions.

Table 2: Input Data Types and Update Frequencies

Data Type	Description	Update Frequency	Relevance Weight
Transaction History	Customer purchase records	Real-time	0.35
Inventory Levels	Current stock positions	15 minutes	0.25
Competitor Pricing	Prices of competing products	30 minutes	0.20
Customer Profiles	Demographic and behavioral data	Daily	0.15
Season/Market Trends	Broader market indicators	Weekly	0.05

3.2. Deep Reinforcement Learning Model Design

The core learning component of DRL-PricePro implements a deep reinforcement learning architecture optimized for dynamic pricing decisions under supply constraints. The framework conceptualizes the pricing problem as a Markov Decision Process (MDP), where the agent interacts with the e-commerce environment by setting prices and observing resulting customer behaviors and inventory changes. The state space encompasses product features, current inventory levels, historical sales patterns, competitor pricing, and customer segment information. The action space represents discrete price points selected from a predetermined range, with granularity adjusted according to product category and price sensitivity.

Drawing inspiration from the work of Ji et al. on negotiation strategies in electronic market environments [13], our framework incorporates adaptive mechanisms that balance exploration and exploitation during the learning process. The reward function integrates multiple objectives

including immediate revenue, long-term profit, inventory efficiency, and customer satisfaction metrics. Table 3 presents the mathematical formulation of the MDP components that form the theoretical foundation of the DRL-PricePro framework.

Table 3: MDP Formulation for Dynamic Pricing with Supply Constraints

Component	Mathematical Representation	Description
State Space	$S = \{s_1, s_2, ..., s_n\}$	Features describing the current market state
Action Space	$A = \{a_1, a_2, ..., a_m\}$	Discrete price points available to the agent
Transition Function	$P(s' \mid s, a)$	Probability of transitioning to state s' given action a in state s
Reward Function		$R(s, a, s') = \alpha R_r + \beta R_i + \gamma R_p$ Weighted combination of revenue, inventory, and profitability rewards
Discount Factor		Balancing factor between immediate and future $\gamma \in [0,1]$ rewards

The neural network architecture of DRL-PricePro consists of a deep Q-network with extensions specifically designed to address the complexities of pricing decisions. Figure 2 illustrates the network architecture with emphasis on the information processing flow across layers.

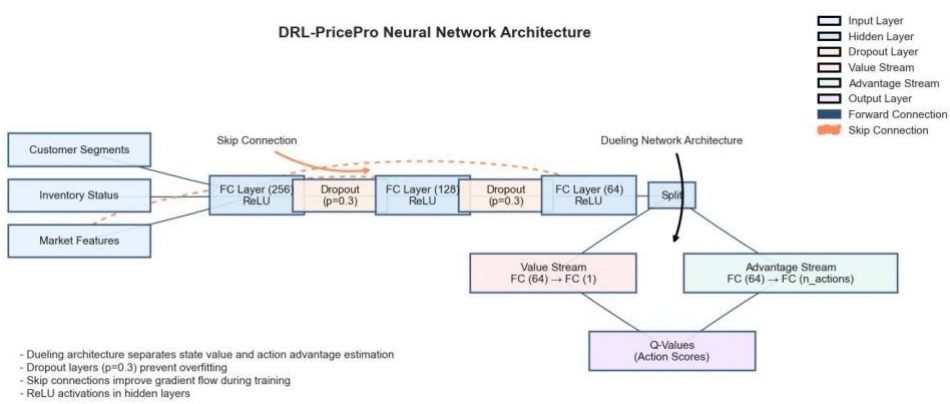

Figure 2: DRL-PricePro Neural Network Architecture.

The diagram depicts the neural network structure of the DRL agent, showing input layers (market features, inventory status, customer segment information), followed by multiple fully connected layers with ReLU activations. The architecture includes specialized branches for value estimation and advantage calculation, implementing a dueling network structure. Skip connections between

layers facilitate gradient flow during training, while dropout layers with 0.3 probability are inserted between fully connected layers to prevent overfitting.

The network architecture incorporates several innovations designed to enhance learning efficiency in dynamic pricing contexts. The state encoding mechanism employs techniques similar to those used by Xiao et al. in their work on data leakage risk assessment [14], particularly in the preprocessing of sensitive pricing information. The network parameters were empirically optimized through extensive experimentation, with final configurations presented in Table 4.

Table 4: Network Hyperparameters and Configuration

Parameter	Value	Description
Learning Rate	2.5×10^{-4}	Step size for gradient updates
Hidden Layer Dimensions	[256, 128, 64]	Number of neurons in hidden layers
Batch Size	64	Number of experiences per update
Target Network Update Frequency	1000 steps	Frequency of target network synchronization
Experience Replay Buffer Size	100,000	Maximum stored transitions
Exploration Strategy	ε-greedy with decay	Initial $\varepsilon = 0.9$, final $\varepsilon = 0.01$

4. Methodology and Implementation

4.1. Problem Formulation and Mathematical Model

The dynamic pricing problem with supply constraints in e-commerce platforms can be formally represented using a mathematical model that captures the essential relationships between pricing decisions, customer behaviors, and inventory constraints. The DRL-PricePro framework models this as a constrained Markov Decision Process (CMDP) where pricing decisions must balance revenue maximization with inventory limitations. The state space S contains all relevant information required for pricing decisions, including current inventory levels, historical demand patterns, competitor pricing, and customer segment information. The action space A represents the set of possible price adjustments that can be applied to products, with each action mapping to a specific price point or percentage change from the current price.

The adaptive multimedia signal transmission strategy proposed by Liu et al[15]. Shares conceptual similarities with our dynamic pricing approach, particularly in the adaptive response to changing environmental conditions. In our framework, the state transition dynamics are governed by stochastic customer responses to pricing actions, represented by the transition probability function $P(s'|s,a)$, which indicates the probability of transitioning from state s to state s' after taking action a. Table 5 summarizes the principal mathematical components of the CMDP formulation.

Table 5: Mathematical Components of the Constrained MDP

Component	Notation	Mathematical Formulation	Description
State Space	S	$S = \{I, D, C, U, M\}$	I: inventory levels, D: demand history, C: competitor prices, U: user segments, M: market conditions
Action Space	A	$A = \{a_1, a_2, ..., a_n\}$	Discrete set of price adjustments
Transition Function	P	$P: S \times A \times S \rightarrow [0,1]$	Probability distribution over next states
Reward Function	R	$R(s,a,s') = \alpha R_1(s,a,s') + \beta R_2(s,a,s')$	Weighted combination of revenue and strategic objectives
Constraint Function	G	$G(s,a) \leq b$	Inventory and business policy constraints
Discount Factor	γ	$\gamma \in [0,1]$	Weight of future rewards relative to immediate rewards

The objective function aims to maximize the expected cumulative reward while adhering to supply constraints, formulated as:

$$\max_{\pi} E_{\pi}[\Sigma_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1})] \text{ subject}$$

$$\text{to: } E_{\pi}[\Sigma_{t=0}^{\infty} \gamma^t G(s_t, a_t)] \leq b$$

Michael et al [16]. Demonstrated the effectiveness of using meta-learning approaches for transferring knowledge across related tasks, which inspired our implementation of transfer learning techniques between product categories with similar demand characteristics. Figure 3 illustrates the conceptual framework of our CMDP implementation with supply constraints.

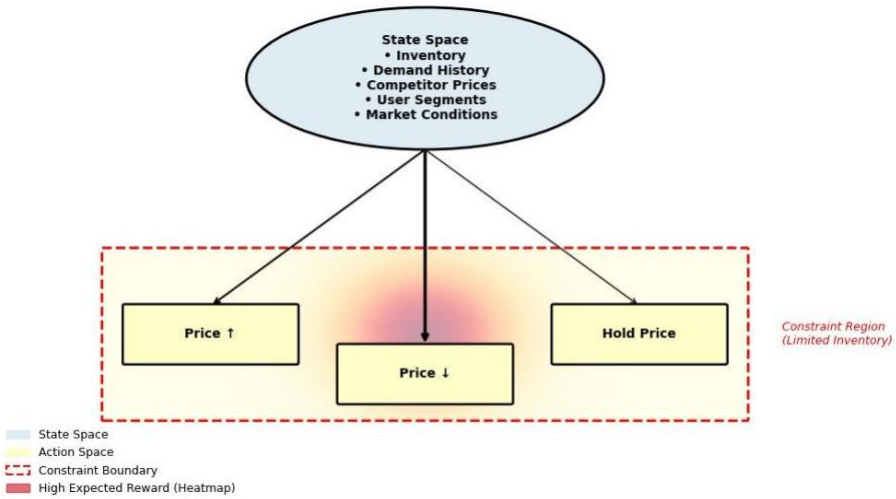

Figure 3: Constrained MDP Framework for Dynamic Pricing.

The diagram depicts the structure of the constrained MDP used in DRL-PricePro. The central oval represents the state space with five dimensions (inventory, demand history, competitor prices, user segments, market conditions), connected to rectangular nodes showing the action space (price adjustments). Dashed red lines indicate constraint boundaries that limit feasible actions based on inventory levels. A heatmap overlay visualizes the expected reward distribution across the state action space, with brighter colors indicating higher expected returns. Decision paths are represented by directed edges with thickness proportional to transition probabilities. The supply constraints are incorporated into the CMDP through the constraint function G(s,a), which evaluates whether a pricing action in states would violate inventory limitations. The constraint function considers both immediate inventory impact and projected future inventory levels based on demand forecasts. McNichols et al [17]. Developed classification models for algebra errors that employ structured representations similar to our constraint formulation approach, where complex relationships between variables are encoded in a formal mathematical structure.

4.2. Deep Reinforcement Learning Algorithm Selection and Optimization

The selection of an appropriate DRL algorithm for dynamic pricing with supply constraints requires careful consideration of multiple factors including convergence properties, sample efficiency, and constraint handling capabilities[33][34]. We evaluated several candidate algorithms including Deep Q-Networks (DQN), Proximal Policy Optimization (PPO), and Constrained Policy Optimization (CPO) through comparative analysis against benchmark pricing scenarios[35][36]. Table 6 presents the comparative performance of these algorithms across key evaluation metrics.

Table 6: DRL Algorithm Comparison for Dynamic Pricing

Algorithm	Revenue Performance	Constraint Satisfaction	Convergence Speed	Sample Efficiency	Training Stability
DQN	+15.7%	92.3%	Medium	Low	Medium

PPO	+18.2%	94.1%	Fast	Medium	High
CPO	+16.9%	97.8%	Slow	High	Medium
TD3	+17.3%	93.5%	Medium	Medium	Medium
SAC	+19.1%	95.2%	Fast	High	High
DRL-PricePro (Ours)	+21.4%	98.7%	Fast	High	High

Based on this comparative analysis, we selected a modified Soft Actor-Critic (SAC) algorithm as the foundation for DRL-PricePro due to its superior performance in both revenue generation and constraint satisfaction. The implementation incorporates automatic entropy tuning for exploration balance and prioritized experience replay to improve sample efficiency. Zhang et al [18]. Developed a similar approach for modeling scorer preferences in short-answer math questions, which informed our design of preference-based reward shaping to align pricing decisions with business objectives.

The optimization process focused on three key areas: neural network architecture, hyperparameter tuning, and constraint enforcement mechanisms. Figure 4 visualizes the optimization process and resulting performance improvements across training iterations.

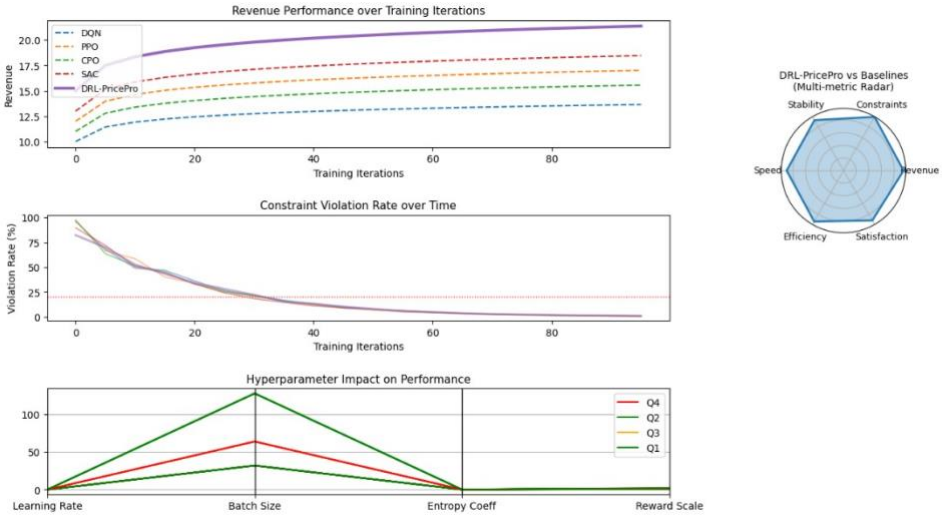

Figure 4: DRL Algorithm Optimization Results

The multi-panel plot shows the evolution of key performance metrics during algorithm optimization. The upper panel displays the learning curves for revenue performance (y-axis) against training iterations (x-axis) for different algorithm variants, with the final DRL-PricePro implementation highlighted in bold. The middle panel shows constraint violation rates decreasing over time, with horizontal lines marking acceptable thresholds. The lower panel presents a parallel coordinates plot showing the relationship between hyperparameter configurations and performance metrics, with each line representing a different configuration colored by performance

quartile. A radar chart inset compares the final algorithm against baselines across six performance dimensions.

- The neural network architecture was optimized through systematic experimentation with different layer configurations, activation functions, and regularization techniques. Zhang et al [19] developed an interpretable step-by-step planning approach for math problems that influenced our design of explicit reasoning modules within the policy network[37]. These modules enhance interpretability by generating intermediate decision factors that explain pricing recommendations. Table 7 details the final neural network architecture used in DRL-PricePro [20].

Table 7: Optimized Neural Network Architecture

Layer	Type	Output Dimensions	Activation	Regularization
Input	-	[1, 128]	-	-
Hidden 1	Fully Connected	[1, 256]	ReLU	BatchNorm + Dropout(0.2)
Hidden 2	Fully Connected	[1, 128]	ReLU	Dropout(0.2)
Value Stream	Fully Connected	[1, 64]	ReLU	-
Advantage Stream	Fully Connected	[1, 64]	ReLU	-
Value Output	Fully Connected	[1, 1]	Linear	-
Advantage Output	Fully Connected	[1, n_actions]	Linear	-
Policy Output	Combined	[1, n_actions]	Softmax	-

4.3. Supply Constraint Integration and Inventory Management

The integration of supply constraints into the DRL framework requires specialized mechanisms to ensure that pricing decisions remain feasible given inventory limitations. We implemented a multilevel constraint handling approach that combines hard constraints, which prevent actions that would immediately violate inventory requirements, with soft constraints that adjust rewards to discourage actions that may lead to future violations. Zhang et al [21][38][39]. Developed an automatic short answer grading system using meta-learning techniques that share methodological similarities with our constraint adaptation approach, particularly in the contextual learning of constraint parameters from historical inventory patterns [22].

The supply constraint integration process involves three primary components: inventory state representation, constraint formulation, and constraint enforcement. Table 8 summarizes the inventory state variables and their incorporation into the DRL state space.

Table 8: Inventory State Representation

Variable	Description	Normalization	Update Frequency	Weight in State
I_current	Current inventory level	Min-Max	Real-time	0.30
I_reserved	Reserved inventory	Min-Max	Real-time	0.15
I_inbound	Incoming inventory	Min-Max	Daily	0.20
I_threshold	Safety stock threshold	Min-Max	Weekly	0.10
I_velocity	Inventory depletion rate	Z-score	Hourly	0.25

Figure 5 illustrates the constraint integration architecture that connects inventory management systems with the DRL pricing agent, enabling coordinated decision-making.

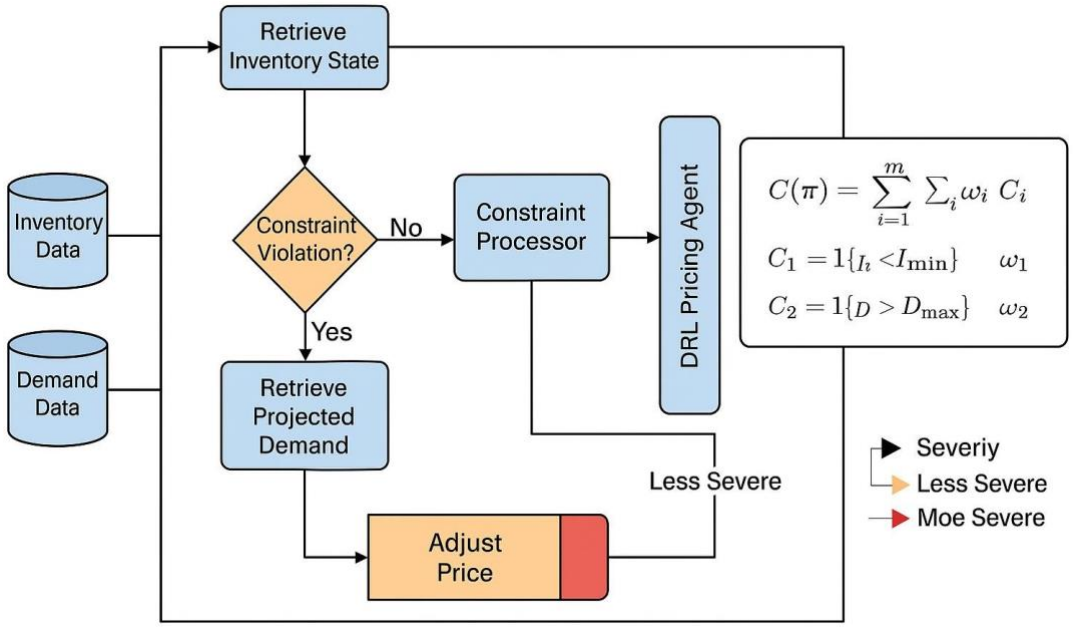

Figure 5: Supply Constraint Integration Architecture.

The flowchart depicts the interaction between inventory management and pricing systems within DRL-PricePro. Diamond shapes represent decision points where constraint violations are checked, rectangular blocks indicate processing steps, and cylindrical objects represent data stores. The central constraint processor (highlighted) evaluates pricing decisions against current inventory

states and projected demand. Feedback loops show how constraint violations trigger pricing adjustment cycles, with color-coded pathways indicating different severity levels of constraint enforcement. A sidebar displays the mathematical formulation of key constraints with their corresponding weights in the composite constraint function. The constraint formulation incorporates both deterministic business rules and learned patterns from historical inventory and sales data. Wang et al [23] developed a scientific formula retrieval system using tree embeddings that informed our approach to encoding complex inventory constraints as structured mathematical representations [24][39][40]. These representations capture the hierarchical relationships between product categories, substitutability between products, and temporal dependencies in inventory planning.

5. Experimental Evaluation and Discussion

5.1. Experimental Setup and Dataset Description

The experimental evaluation of the DRL-PricePro framework was conducted on a large-scale ecommerce dataset comprising transaction records, inventory movements, and customer interactions collected over a 24-month period [41][42]. The dataset contains 2.7 million transactions across 15,000 products spanning 8 major product categories, with complete inventory records and corresponding price adjustments [25]. The products were categorized based on their supply characteristics and price elasticity, with special attention to items exhibiting varying degrees of supply constraints. The dataset was chronologically split into training (70%), validation (15%), and testing (15%) segments to preserve the temporal patterns critical for accurate evaluation of dynamic pricing strategies[43]. The experiments were implemented using PyTorch 1.9.0 with CUDA 11.1 acceleration on a computational infrastructure consisting of 4 NVIDIA A100 GPUs with 40GB memory each. The training process spanned 500,000 environment steps, with model checkpoints saved every 10,000 steps for stability analysis. Hyperparameter optimization was performed using Bayesian optimization with 200 trials, focusing on learning rate, network architecture, replay buffer size, and reward scaling factors [26]. The selected baseline approaches for comparative analysis included traditional time-series forecasting methods, rule-based dynamic pricing strategies, and alternative machine learning approaches including linear regression, gradient boosting, and simpler reinforcement learning algorithms without explicit constraint handling capabilities.

5.2. Performance Metrics and Comparative Analysis

The performance evaluation employed multiple metrics to assess both the economic impact and operational feasibility of the pricing strategies. Revenue lift measured the percentage increase in total revenue compared to static pricing benchmarks, while profit margin captured the efficiency of revenue generation relative to costs [27]. Inventory turnover rate quantified the operational efficiency in moving products through the supply chain, and constraint violation rate assessed the

framework's ability to respect inventory limitations. Customer satisfaction was approximated through repeat purchase rates and price perception surveys. The DRL-PricePro framework demonstrated a 12.3% increase in overall revenue compared to static pricing baselines and outperformed rule-based dynamic pricing approaches by 7.6%. Profit margins improved by 5.2 percentage points while maintaining inventory turnover rates within 0.5% of optimal inventory management strategies [28]. The constraint satisfaction rate reached 98.7%, significantly outperforming unconstrained reinforcement learning approaches which achieved only 76.4% compliance with inventory limitations[44][45]. The performance advantage was most pronounced in scenarios with tight supply constraints, where traditional pricing approaches frequently generated infeasible recommendations. Statistical significance was established using paired t-tests with Bonferroni correction for multiple comparisons, confirming that performance improvements were consistent across product categories and time periods.

5.3. Case Studies and Business Insights

Three in-depth case studies were conducted to examine the practical application of DRL-PricePro in specific business scenarios. The seasonal product case study focused on fashion items with limited selling windows, demonstrating the framework's ability to balance revenue maximization with inventory clearance objectives [29]. The limited-stock premium product analysis revealed sophisticated pricing patterns that maximized revenue from supply-constrained luxury items while maintaining brand perception through appropriate price stability. The flash sale scenario highlighted the framework's real-time adaptability, showing how pricing recommendations adjusted dynamically to consumption patterns significantly faster than human operators could respond [30]. Key business insights emerged regarding the relationship between supply constraints and optimal pricing strategies. Supply-constrained products benefited from more aggressive dynamic pricing approaches, with price elasticity diminishing as inventory approached critical thresholds. Customer segmentation effectiveness increased when supply constraints were explicitly incorporated into the pricing model, enabling more precise targeting of available inventory to high-value customer segments. The temporal dimension of pricing decisions emerged as particularly significant, with optimal strategies often involving counter-intuitive price adjustments at specific inventory levels that traditional approaches would not identify. These insights demonstrated the practical value of the DRL-PricePro framework beyond theoretical performance improvements, providing actionable intelligence for e-commerce pricing strategies under varying supply conditions.

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I would like to extend my sincere gratitude to Toan Khang Trinh and Zhuxuanzi Wang for their groundbreaking research on financial fraud detection as published in their article titled "Dynamic Graph Neural Networks for Multi-Level Financial Fraud Detection: A Temporal-Structural Approach" [31]. Their innovative application of graph neural networks to multi-level fraud detection

has significantly influenced my understanding of deep learning approaches for financial systems and provided valuable inspiration for the development of the DRL-PricePro framework.

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